

Dendrites

Projections of neuron; collect nerve impulses from other neurons or sensory nerve endings

Neuron cell body

Main part of the neuron, containing the nucleus and cell components

Axon

Main nerve fibre of the neuron, conveying impulses away from the cell body

Neurofibril node

Also called node of Ranvier; portion of axon not covered by myelin

Myelin sheath

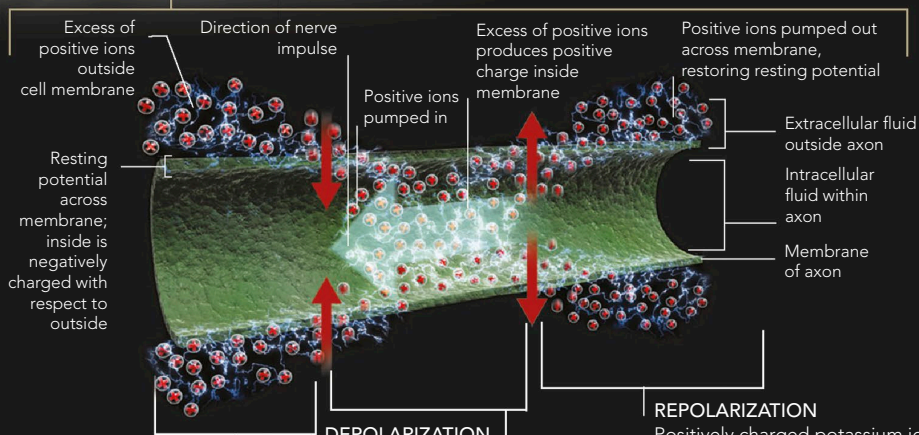
Also called neurilemma or Schwann sheath; spiralling structure of fatty myelin that helps to speed an impulse and prevent it from fading or leaking

Schwann cell

Sheet-like cell that grows around a portion of axon (fibre) to form the myelin sheath

IMPULSE MOVEMENT WITHIN A NERVE CELL

A nerve impulse is based chiefly on movement of positively charged sodium and potassium ions through the neuron's cell membrane. Impulses travel at speeds of between 1 and 120m/s (3–400ft/s), depending on the type of nerve. Movement is faster in myelin-coated axons.

**RESTING POTENTIAL**

There are more positive ions outside the cell and more negative ions inside, producing a "resting potential" of -70 millivolts.

DEPOLARIZATION

Positive ions rush in through ion channels. The membrane is first depolarized, then its polarity is reversed, resulting in an "action potential" of $+30$ millivolts inside.

REPOLARIZATION

Positively charged potassium ions flow in the opposite direction, restoring the charge balance. This stimulates an adjacent area of membrane, and the next, and the impulse moves along like a wave.